The spaces L^p: Hölder and Minkowski inequalities

REAL VARIABLES I

Fall 2026

Up to now, we measured the size of an integrable function f by evaluating $\int_X |f| \, \mathrm{d}\mu$. The dominated convergence theorem established conditions under which $\int_X |f_n - f| \, \mathrm{d}\mu \to 0$. For an exponent $p > 0$, the integrals $\int_X |f|^p \, \mathrm{d}\mu$ are a family of ways of measuring the size of a function f. When $p > 1$ the size amplifies values where $|f(x)|$ is large, whereas exponents $p < 1$ amplify values where $|f(x)|$ is small. These notes introduce the function spaces $L^p(X, \mu)$ of measurable functions for which $\int_X |f|^p \, \mathrm{d}\mu < +\infty$, and investigate whether the quantity

$$\|f\|_p \coloneqq \left(\int_X |f|^p \, \mathrm{d}\mu \right)^{1/p}$$

satisfies the properties of a norm, with particular focus on the triangle inequality.

For $p = 1$, the triangle inequality $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$ follows by integrating the pointwise inequality $|f(x) + g(x)| \leq |f(x)| + |g(x)|$. When $p > 1$, the exponent $1/p$ does not commute with integration, and the pointwise inequality $|a + b|^p \leq |a|^p + |b|^p$ fails for positive numbers. To establish the triangle inequality in this setting, we proceed in three stages.

First, the concavity of the natural logarithm yields Young's inequality, which bounds a product of two non-negative numbers by a weighted sum of powers. Second, integrating Young's inequality after normalizing functions by their respective norms produces Hölder's inequality, which bounds the integral of a product $\int_X |fg| \, \mathrm{d}\mu$ by $\|f\|_p \, \|g\|_{p'}$ for a conjugate exponent p' satisfying $1/p + 1/p' = 1$. Third, applying Hölder's inequality to the components of $|f + g|^p$ establishes Minkowski's inequality, which is the triangle inequality $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Next, we breifly discuss L^p spaces with exponents $p \in (0, 1)$, where the triangle inequality fails but a quasinorm inequality holds.

Finally, we compare the spaces L^p for different exponents on finite and infinite measure spaces. We also interpret Hölder's inequality as a result of duality of spaces L^p and $L^{p'}$: $L^{p'}$ are naturally bounded linear functionals on L^p.

After studying these notes, you will be able to do the following.

- Define $L^p(X, \mu)$ as a space of almost-everywhere equivalence classes and explain why identifying functions that agree almost everywhere is necessary for $\|\cdot\|_p$ to satisfy the point-separation axiom of a norm.
- Determine the exponents $p \in (0, +\infty)$ for which the power function $x \mapsto x^{-a}$ belongs to $L^p(0, 1]$ and to $L^p[1, +\infty)$.
- Prove Young's inequality from the concavity of the logarithm, and deduce Hölder's inequality via homogenization.
- Prove Minkowski's inequality and verify that $\|\cdot\|_p$ defines a norm on $L^p(X, \mu)$ for every $p \in [1, +\infty)$.
- Prove that $\|\cdot\|_p$ is a quasinorm on $L^p(X, \mu)$ for every $p \in (0, +\infty)$, and demonstrate the failure of the triangle inequality when $p \in (0, 1)$.
- Prove the inclusion $L^q(X, \mu) \subseteq L^p(X, \mu)$ for $0 < p < q$ when $\mu(X) < +\infty$, and construct functions showing that this inclusion fails on $\mathbb{R}$ with Lebesgue measure.

- Apply Hölder's inequality to show that each element of $L^{p'}(X,\mu)$ defines a bounded linear functional on $L^p(X,\mu)$ whose operator norm equals the $L^{p'}$ norm.

Throughout these notes, $(X,\mathcal{A},\mu)$ denotes a measure space, and functions are measurable functions $f\colon X\to\mathbb{R}$. On subsets of $\mathbb{R}$, the symbol $\mathcal{L}$ denotes Lebesgue measure.

11.1 Measuring size with p-th powers

The integral $\int_X |f|^p \,\mathrm{d}\mu$ measures the size of a measurable function $f\colon X\to\mathbb{R}$. Varying the exponent p changes which functions have finite integrals. Power functions on subsets of the real line provide a concrete illustration of this dependence and illustrate the distinct roles of singularities and decay at infinity.

Example 11.1 (Power functions near zero and near infinity). For an exponent $s>0$, the power function $x\mapsto x^{-s}$ is continuous on every compact interval in $(0,+\infty)$. Because continuous functions on compact intervals are Riemann integrable and the Riemann integral agrees with the Lebesgue integral (Note 23, Theorem 23.1), we compute for $s\neq 1$ and $n\geq 2$:

$$\int_{[1/n,1]} x^{-s}\,\mathrm{d}\mathcal{L}(x)=\frac{1-n^{s-1}}{1-s},\qquad \int_{[1,n]} x^{-s}\,\mathrm{d}\mathcal{L}(x)=\frac{n^{1-s}-1}{1-s},$$

while for $s=1$ both integrals evaluate to $\log n$. As $n\to\infty$, the sequence of functions $x^{-s}\mathbb{1}_{[1/n,1]}$ increases pointwise to $x^{-s}\mathbb{1}_{(0,1]}$, and the sequence $x^{-s}\mathbb{1}_{[1,n]}$ increases pointwise to $x^{-s}\mathbb{1}_{[1,+\infty)}$. The monotone convergence theorem therefore implies

$$\int_{(0,1]} x^{-s}\,\mathrm{d}\mathcal{L}(x)<+\infty \iff s<1,\qquad \int_{[1,+\infty)} x^{-s}\,\mathrm{d}\mathcal{L}(x)<+\infty \iff s>1. \tag{11.1}$$

For a fixed parameter $a>0$, applying the two integrability equivalences in (11.1) with $s=ap$ to the p-th power $|x^{-a}|^p=x^{-ap}$ yields

$$\int_{(0,1]} (x^{-a})^p\,\mathrm{d}\mathcal{L}(x)<+\infty \iff p<\frac{1}{a},\qquad \int_{[1,+\infty)} (x^{-a})^p\,\mathrm{d}\mathcal{L}(x)<+\infty \iff p>\frac{1}{a}.$$

This example demonstrates two distinct mechanisms that can cause an integral to diverge. Near $x=0$, the function x^{-a} is unbounded, so the integrability of $|x^{-a}|^p$ requires $p<1/a$ to prevent excessive concentration of mass on small intervals $(0,\delta)$. Near $+\infty$, the function x^{-a} decays slowly, and the integrability of $|x^{-a}|^p$ requires $p>1/a$ to ensure sufficient decay across the set of infinite measure $[1,+\infty)$. These mechanisms correspond to the phenomena of mass concentration and escape of mass to infinity analyzed in the study of integral limit theorems (Note 22).

11.1.1 The space $\mathcal{L}^p$ and the L^p norm

Definition 11.2 (p-integrable functions). For $p\in[1,+\infty)$, the set $\mathcal{L}^p(X,\mu)$ consists of the measurable functions $f\colon X\to\mathbb{R}$ such that

$$\int_X |f|^p\,\mathrm{d}\mu<+\infty.$$

For $f \in \mathcal{L}^p(X,\mu)$, the L^p **norm** of f is

$$\|f\|_p \coloneqq \left(\int_X |f|^p \, \mathrm{d}\mu\right)^{1/p}. \tag{11.2}$$

The symbol $\|f\|_p$ is read "the L^p norm of f." In the sections that follow, we verify that $\|\cdot\|_p$ satisfies the axioms of a norm on a suitable quotient space. When the underlying measure space is clear from context, we write $\mathcal{L}^p$ in place of $\mathcal{L}^p(X,\mu)$. For $p = 1$, the set $\mathcal{L}^1(X,\mu)$ consists of the Lebesgue integrable functions, and $\|f\|_1 = \int_X |f| \, \mathrm{d}\mu$. For the counting measure on $\mathbb{N}$, the integral coincides with a series, and $\mathcal{L}^p(\mathbb{N})$ is the space of real sequences $(a_k)_{k\in\mathbb{N}}$ satisfying $\sum_{k=1}^{+\infty} |a_k|^p < +\infty$; this sequence space is denoted ℓ^p.

The power $1/p$ in (11.2) ensures absolute homogeneity of degree 1: for every $\lambda \in \mathbb{R}$ and $f \in \mathcal{L}^p(X,\mu)$,

$$\|\lambda f\|_p = \left(|\lambda|^p \int_X |f|^p \, \mathrm{d}\mu\right)^{1/p} = |\lambda| \, \|f\|_p. \tag{11.3}$$

Scaling f by a factor of λ scales $\|f\|_p$ by $|\lambda|$; without the root $1/p$, scaling f by 2 would multiply the integral by 2^p rather than by 2.

The set $\mathcal{L}^p(X,\mu)$ is a real vector space. Indeed, for every $\lambda \in \mathbb{R}$ and $f \in \mathcal{L}^p(X,\mu)$, the scalar multiple λf is measurable and satisfies $\|\lambda f\|_p = |\lambda| \, \|f\|_p < +\infty$ by (11.3), so $\lambda f \in \mathcal{L}^p(X,\mu)$. For $f, g \in \mathcal{L}^p(X,\mu)$, the sum $f + g$ is measurable. At every point $x \in X$, the triangle inequality in $\mathbb{R}$ gives $|f(x) + g(x)| \le |f(x)| + |g(x)| \le 2\max\{|f(x)|, |g(x)|\}$. Because the function $t \mapsto t^p$ is non-decreasing on $[0, +\infty)$,

$$|f(x) + g(x)|^p \le (2\max\{|f(x)|, |g(x)|\})^p \le 2^p(|f(x)|^p + |g(x)|^p). \tag{11.4}$$

Integrating (11.4) over X with respect to μ yields

$$\int_X |f + g|^p \, \mathrm{d}\mu \le 2^p \left(\int_X |f|^p \, \mathrm{d}\mu + \int_X |g|^p \, \mathrm{d}\mu\right) = 2^p(\|f\|_p^p + \|g\|_p^p) < +\infty.$$

Thus $f + g \in \mathcal{L}^p(X,\mu)$. Because the estimates (11.3) and (11.4) require only $p > 0$, the set $\mathcal{L}^p(X,\mu)$ is a real vector space for every exponent $p \in (0, +\infty)$.

Taking the p-th root of this integral bound yields

$$\|f + g\|_p \le 2(\|f\|_p^p + \|g\|_p^p)^{1/p}.$$

When $p \ge 1$, the numerical inequality $(a^p + b^p)^{1/p} \le a + b$ holds for all $a, b \ge 0$, which gives $\|f+g\|_p \le 2(\|f\|_p + \|g\|_p)$. When $p \in (0,1)$, the estimate $a^p + b^p \le 2\max\{a^p, b^p\} \le 2(a+b)^p$ gives $(a^p + b^p)^{1/p} \le 2^{1/p}(a+b)$, which yields $\|f+g\|_p \le 2^{1+1/p}(\|f\|_p + \|g\|_p)$. In either case, the quantity $\|f+g\|_p$ satisfies a weakened version of the triangle inequality with a multiplicative constant on the right-hand side. Such an estimate is called a **quasi-triangle inequality**.

Definition 11.3 (Quasinorm and quasinormed space). A **quasinorm** on a real vector space V is a function $\|\cdot\| \colon V \to [0, +\infty)$ for which there exists a constant $C \ge 1$ such that, for all $u, v \in V$ and $\lambda \in \mathbb{R}$,

$$\|v\| = 0 \iff v = 0, \qquad \|\lambda v\| = |\lambda| \, \|v\|, \qquad \|u + v\| \le C(\|u\| + \|v\|).$$

A vector space equipped with a quasinorm is called a **quasinormed space**. When $C = 1$, the quasi-triangle inequality becomes the triangle inequality $\|u + v\| \le \|u\| + \|v\|$, and $\|\cdot\|$ is a **norm**.

Once functions that agree almost everywhere are identified (Section 11.1.2), the functional $\|\cdot\|_p$ satisfies the separation axiom, and the estimate above shows that $L^p(X,\mu)$ is a quasinormed space for every exponent

$p \in (0, +\infty)$. However, for exponents $p \geq 1$, a central goal of the sections that follow is to establish that $L^p(X, \mu)$ is a genuine **normed space**: that is, the triangle inequality

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

holds with the optimal constant $C = 1$. Obtaining this optimal bound for $p \geq 1$ is the content of Minkowski's inequality (Theorem 11.11), which will be proved via Hölder's inequality for products in Section 11.2 and Section 11.3. In contrast, for exponents $p \in (0, 1)$, we will show in Section 11.5 that the triangle inequality with $C = 1$ fails, so $L^p(X, \mu)$ remains strictly a quasinormed space, with optimal quasinorm constant $C_p = 2^{1/p-1}$.

11.1.2 Almost-everywhere equivalence

In a normed vector space, the norm vanishes if and only if the vector is zero. On the function space $\mathcal{L}^p(X, \mu)$, however, the quantity $\|f\|_p$ can vanish for functions that are not identically zero. For a non-negative measurable function h, the integral $\int_X h \, \mathrm{d}\mu$ vanishes if and only if $h = 0$ almost everywhere (Note 21, Proposition 21.1). Applying this property to $h = |f|^p$ yields

$$\|f\|_p = 0 \iff \int_X |f|^p \, \mathrm{d}\mu = 0 \iff |f|^p = 0 \text{ almost everywhere} \iff f = 0 \text{ almost everywhere.}$$

For example, on $\mathbb{R}$ with Lebesgue measure, the indicator function $\mathbb{1}_{\mathbb{Q}}$ is non-zero, yet $\|\mathbb{1}_{\mathbb{Q}}\|_p = \mathcal{L}(\mathbb{Q})^{1/p} = 0$. More generally, if two functions $f, g \in \mathcal{L}^p(X, \mu)$ satisfy $f = g$ almost everywhere, then $|f - g|^p = 0$ almost everywhere, so $\|f - g\|_p = 0$. The integral cannot distinguish between functions that differ only on a set of measure zero. To obtain a space on which $\|\cdot\|_p$ satisfies the separation axiom of a norm, we identify functions that agree almost everywhere.

Definition 11.4 (Almost-everywhere equivalence). Two measurable functions $f, g \colon X \to \mathbb{R}$ are **equivalent almost everywhere**, written $f \sim g$, if

$$\mu(\{x \in X : f(x) \neq g(x)\}) = 0.$$

The set $\{x \in X : f(x) \neq g(x)\} = \{x \in X : (f - g)(x) \neq 0\}$ is measurable because $f - g$ is measurable, so Definition 11.4 is well-posed. The relation $\sim$ is an equivalence relation on the space of measurable functions. Reflexivity and symmetry follow from $\{f \neq f\} = \emptyset$ and $\{f \neq g\} = \{g \neq f\}$. To verify transitivity, suppose that $f \sim g$ and $g \sim h$. The inclusion $\{x \in X : f(x) \neq h(x)\} \subseteq \{x \in X : f(x) \neq g(x)\} \cup \{x \in X : g(x) \neq h(x)\}$ and the subadditivity of μ give

$$\mu(\{f \neq h\}) \leq \mu(\{f \neq g\}) + \mu(\{g \neq h\}) = 0 + 0 = 0,$$

which proves that $f \sim h$.

The relation $\sim$ respects the vector operations and the L^p integral. For addition, the inclusion

$$\{x \in X : (f + g)(x) \neq (\tilde{f} + \tilde{g})(x)\} \subseteq \{x \in X : f(x) \neq \tilde{f}(x)\} \cup \{x \in X : g(x) \neq \tilde{g}(x)\}$$

shows that $f + g \sim \tilde{f} + \tilde{g}$ whenever $f \sim \tilde{f}$ and $g \sim \tilde{g}$. Similarly, $\{\lambda f \neq \lambda \tilde{f}\} \subseteq \{f \neq \tilde{f}\}$ implies that $\lambda f \sim \lambda \tilde{f}$ for every $\lambda \in \mathbb{R}$. Finally, if $f \sim \tilde{f}$, then $|f|^p = |\tilde{f}|^p$ almost everywhere. Because functions that agree almost everywhere have equal integrals (Note 21), it follows that $\int_X |f|^p \, \mathrm{d}\mu = \int_X |\tilde{f}|^p \, \mathrm{d}\mu$. In particular, $f \in \mathcal{L}^p(X, \mu)$ if and only if $\tilde{f} \in \mathcal{L}^p(X, \mu)$, and in that case $\|f\|_p = \|\tilde{f}\|_p$.

Definition 11.5 (The space L^p). For $p \in [1, +\infty)$, the space $L^p(X, \mu)$ is the set of equivalence classes

$$[f] \coloneqq \{g \in \mathcal{L}^p(X, \mu) : g \sim f\}, \qquad f \in \mathcal{L}^p(X, \mu),$$

with vector addition $[f] + [g] \coloneqq [f + g]$, scalar multiplication $\lambda[f] \coloneqq [\lambda f]$, and the functional $\|[f]\|_p \coloneqq \|f\|_p$.

Because almost-everywhere equivalence preserves vector addition, scalar multiplication, and the integral $\int_X |f|^p \, \mathrm{d}\mu$, the operations $[f] + [g]$ and $\lambda[f]$ and the quantity $\|[f]\|_p$ are well defined independently of the chosen representative $f \in \mathcal{L}^p(X, \mu)$. The space $L^p(X, \mu)$ is a real vector space whose zero element is the equivalence class $[0]$ of functions vanishing almost everywhere. On this quotient space, the functional $\| \cdot \|_p$ satisfies the separation axiom:

$$\|[f]\|_p = 0 \iff f = 0 \text{ almost everywhere} \iff [f] = [0].$$

Following standard mathematical usage, we write f rather than $[f]$ for an element of $L^p(X, \mu)$ and refer to it as a function, with the understanding that f represents its entire almost-everywhere equivalence class.

Warning (Pointwise values of an element of L^p). An element $f \in L^p(X, \mu)$ does not possess a well-defined pointwise value $f(x_0)$ at a single point x_0 satisfying $\mu(\{x_0\}) = 0$: modifying a representative function at x_0 produces another representative of the same equivalence class. Statements concerning elements of $L^p(X, \mu)$ must therefore be invariant under changes on sets of measure zero. For instance, the condition $f \geq 0$ in $L^p(X, \mu)$ means that $f \geq 0$ almost everywhere, and the indicator $\mathbb{1}_{\mathbb{Q}}$ represents the zero element in $L^p(\mathbb{R}, \mathcal{L})$.

11.1.3 The triangle inequality and the need for product estimates

On the space $L^p(X, \mu)$ with $p \in [1, +\infty)$, the functional $\|\cdot\|_p$ satisfies the separation axiom $\|f\|_p = 0 \iff f = 0$ in $L^p(X, \mu)$ by Definition 11.5 and absolute homogeneity $\|\lambda f\|_p = |\lambda| \, \|f\|_p$ by (11.3). To establish that $\| \cdot \|_p$ is a norm on $L^p(X, \mu)$, it remains to prove the triangle inequality

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p \qquad \text{for all } f, g \in L^p(X, \mu).$$

For $p = 1$, integrating the pointwise triangle inequality $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ over X yields $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$. For $p > 1$, however, the function $t \mapsto t^p$ is strictly convex on $[0, +\infty)$, which prevents a direct pointwise deduction: the inequality $|a + b|^p \leq |a|^p + |b|^p$ fails for non-zero numbers $a, b > 0$ (for instance, $(1 + 1)^p = 2^p > 2 = 1^p + 1^p$). The crude bound (11.4) yields only $\|f + g\|_p \leq 2(\|f\|_p^p + \|g\|_p^p)^{1/p}$, which is strictly larger than $\|f\|_p + \|g\|_p$ whenever $f = g \neq 0$.

The strategy for proving the triangle inequality when $p > 1$ (Minkowski's inequality, Section 11.4) relies on decomposing $|f + g|^p$ into the sum of two products,

$$|f(x) + g(x)|^p \leq |f(x)| \, |f(x) + g(x)|^{p-1} + |g(x)| \, |f(x) + g(x)|^{p-1},$$

and bounding the integral of each product separately (in the first product, the two functions are $|f|$ and $|f + g|^{p-1}$; in the second product, they are $|g|$ and $|f + g|^{p-1}$). Carrying out this estimation requires a general upper bound on the integral of a product of two functions, $\int_X |uv| \, \mathrm{d}\mu$, in terms of the individual norms $\|u\|_p$ and $\|v\|_{p'}$ for a suitable conjugate exponent p'. That bound is Hölder's inequality, which we develop in Section 11.2 and Section 11.3 starting from the concavity of the natural logarithm.

11.2 Concavity and Young's inequality

Young's inequality bounds a product ab of non-negative real numbers by a sum of weighted powers of a and b. The proof relies on the concavity of the natural logarithm.

Definition 11.6 (Concave function). For an interval $I \subseteq \mathbb{R}$, a function $\varphi\colon I \to \mathbb{R}$ is **concave** if

$$\varphi((1-t)x + ty) \geq (1-t)\varphi(x) + t\varphi(y) \qquad \text{for all } x, y \in I \text{ and } t \in [0,1].$$

A function $\psi\colon I \to \mathbb{R}$ is **convex** if $-\psi$ is concave.

As the parameter t varies over $[0,1]$, the combination $(1-t)x + ty$ traverses the line segment between x and y, and the quantity $(1-t)\varphi(x) + t\varphi(y)$ represents the height of the chord joining $(x, \varphi(x))$ to $(y, \varphi(y))$. Concavity expresses the geometric condition that every chord lies on or below the graph of the function, as illustrated in Figure 11.1.

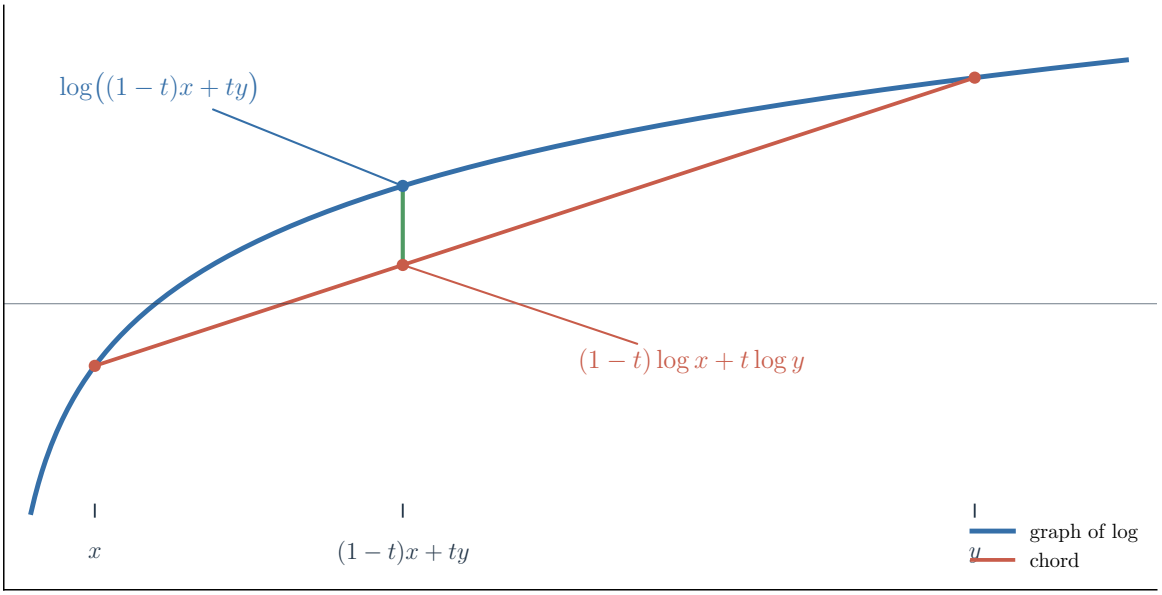


Figure 11.1: The graph of log lies above the chord joining $(x, \log x)$ to $(y, \log y)$. At the point $(1-t)x + ty$, the gap between the graph and the chord is non-negative.

Figure description: The concave graph of the logarithm with a straight chord joining two of its points. A vertical segment at an intermediate point shows the graph value above the chord value.

For twice differentiable functions on an interval, concavity is characterized by the non-positivity of the second derivative.

Lemma 11.7 (Concavity from the second derivative). For an open interval $I \subseteq \mathbb{R}$ and a twice differentiable function $\varphi\colon I \to \mathbb{R}$ with $\varphi'' \leq 0$ on I, the function φ is concave.

Proof of Lemma 11.7. The inequality in Definition 11.6 holds as an equality when $t \in \{0,1\}$ or $x = y$, and it is invariant under exchanging (x,t) with $(y, 1-t)$. We may therefore assume without loss of generality that $x < y$ and $t \in (0,1)$. We define $z \coloneqq (1-t)x + ty$, so that $x < z < y$, with differences $z - x = t(y-x)$ and $y - z = (1-t)(y-x)$. By the mean value theorem, there exist intermediate points $\xi_1 \in (x, z)$ and $\xi_2 \in (z, y)$ such that

$$\frac{\varphi(z) - \varphi(x)}{z - x} = \varphi'(\xi_1), \qquad \frac{\varphi(y) - \varphi(z)}{y - z} = \varphi'(\xi_2).$$

Because $\varphi'' \leq 0$ on I, the first derivative φ' is non-increasing. Since $\xi_1 < z < \xi_2$, we have $\varphi'(\xi_1) \geq \varphi'(\xi_2)$.

Substituting the expressions for $z-x$ and $y-z$ gives

$$\frac{\varphi(z)-\varphi(x)}{t(y-x)} \geq \frac{\varphi(y)-\varphi(z)}{(1-t)(y-x)}.$$

Multiplying both sides by the positive quantity $t(1-t)(y-x)$ yields

$$(1-t)(\varphi(z)-\varphi(x)) \geq t(\varphi(y)-\varphi(z)),$$

which rearranges to $\varphi(z) \geq (1-t)\varphi(x)+t\varphi(y)$. This proves that φ is concave on I. □

On the interval $(0,+\infty)$, the natural logarithm satisfies $\log''(x) = -1/x^2 < 0$, so $\log$ is concave on $(0,+\infty)$ by Lemma 11.7. Similarly, for $r \geq 1$, the power function $s \mapsto s^r$ on $(0,+\infty)$ satisfies $\frac{\mathrm{d}^2}{\mathrm{d}s^2} s^r = r(r-1)s^{r-2} \geq 0$, so $s \mapsto s^r$ is convex on $(0,+\infty)$; this property will be used in the proof of Theorem 11.14.

Young's inequality pairs an exponent p with an associated conjugate exponent.

Definition 11.8 (Conjugate exponent). For $p \in (1,+\infty)$, the **conjugate exponent** of p is the number $p' \in (1,+\infty)$ defined by

$$\frac{1}{p}+\frac{1}{p'} = 1, \qquad \text{equivalently} \qquad p' = \frac{p}{p-1}.$$

The symbol p' is read "p prime" or "the conjugate exponent of p." For example, $2' = 2$, $3' = 3/2$, and $(3/2)' = 3$. The relation $1/p + 1/p' = 1$ is symmetric in p and p', so $(p')' = p$. As p decreases toward 1, the conjugate exponent p' increases toward $+\infty$. Two algebraic identities follow directly from Definition 11.8 and are used throughout these notes:

$$(p-1)p' = p, \qquad \frac{p}{p'} = p-1. \tag{11.5}$$

Theorem 11.9 (Young's inequality). For $p \in (1,+\infty)$ with conjugate exponent p', and for all real numbers $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}. \tag{11.6}$$

Proof of Theorem 11.9. If $a = 0$ or $b = 0$, the left-hand side of (11.6) vanishes and the right-hand side is non-negative. Suppose that $a > 0$ and $b > 0$. The right-hand side of (11.6) is a convex combination of a^p and $b^{p'}$ with weights $1-t = 1/p$ and $t = 1/p'$, which satisfy $(1-t)+t = 1$. Applying the concavity of the natural logarithm (Lemma 11.7) with $x = a^p$, $y = b^{p'}$, and $t = 1/p'$ yields

$$\log\left(\frac{a^p}{p}+\frac{b^{p'}}{p'}\right) \geq \frac{1}{p}\log(a^p)+\frac{1}{p'}\log(b^{p'}) = \log a + \log b = \log(ab).$$

Because the exponential function is strictly increasing on $\mathbb{R}$, applying it to both sides preserves the inequality and establishes (11.6). □

Equality in Young's inequality is characterized by the relation $a^p = b^{p'}$. Indeed, if $a^p = b^{p'}$, then $b = a^{p/p'} = a^{p-1}$ by (11.5), and hence

$$\frac{a^p}{p}+\frac{b^{p'}}{p'} = a^p\left(\frac{1}{p}+\frac{1}{p'}\right) = a^p = a\cdot a^{p-1} = ab.$$

In the special case $p = 2$, the conjugate exponent is $p' = 2$, and Young's inequality reduces to the arithmetic-geometric mean inequality $ab \leq 1/2(a^2+b^2)$, with equality if and only if $a = b$.

11.3 Hölder's inequality

Young's inequality bounds the product of two numbers. We now apply it to estimate the integral of a product of two functions, $\int_X |fg| \, \mathrm{d}\mu$, where $f \in L^p(X, \mu)$ and $g \in L^{p'}(X, \mu)$ for conjugate exponents $p, p' \in (1, +\infty)$. Applying Young's inequality (11.6) pointwise with $a = |f(x)|$ and $b = |g(x)|$ yields

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^{p'}}{p'} \qquad \text{for all } x \in X.$$

Integrating this inequality over X gives

$$\int_X |fg| \, \mathrm{d}\mu \leq \frac{\|f\|_p^p}{p} + \frac{\|g\|_{p'}^{p'}}{p'}. \tag{11.7}$$

While (11.7) provides an upper bound on $\int_X |fg| \, \mathrm{d}\mu$, the two sides of the inequality scale differently under scalar multiplication. For any $\lambda > 0$, replacing f by λf and g by $\lambda^{-1} g$ leaves the product integral $\int_X |(\lambda f)(\lambda^{-1} g)| \, \mathrm{d}\mu = \int_X |fg| \, \mathrm{d}\mu$ unchanged, whereas the right-hand side of (11.7) becomes

$$\frac{\lambda^p}{p}\|f\|_p^p + \frac{\lambda^{-p'}}{p'}\|g\|_{p'}^{p'}.$$

When $\|f\|_p > 0$ and $\|g\|_{p'} > 0$, this expression diverges to $+\infty$ both as $\lambda \to 0$ and as $\lambda \to +\infty$. To obtain an estimate whose scaling behavior matches the bilinearity of the product integral, we normalize f and g to unit norm before integrating. This technique of eliminating scaling mismatches by passing to normalized representatives is called **homogenization**.

Theorem 11.10 (Hölder's inequality). For $p \in (1, +\infty)$ with conjugate exponent p', suppose that $f \in L^p(X, \mu)$ and $g \in L^{p'}(X, \mu)$. Then $fg \in L^1(X, \mu)$ and

$$\int_X |fg| \, \mathrm{d}\mu \leq \|f\|_p \, \|g\|_{p'}. \tag{11.8}$$

Proof of Theorem 11.10. Step 1: Measurability and degenerate cases.
The product fg is measurable because f and g are measurable functions. If $\|f\|_p = 0$, then $f = 0$ almost everywhere by Definition 11.5. Consequently, $fg = 0$ almost everywhere, so $\int_X |fg| \, \mathrm{d}\mu = 0$ and both sides of (11.8) vanish. The same conclusion holds if $\|g\|_{p'} = 0$.

Step 2: Normalization and pointwise estimate.
Suppose that $\|f\|_p > 0$ and $\|g\|_{p'} > 0$. We define the normalized functions

$$F \coloneqq \frac{|f|}{\|f\|_p}, \qquad G \coloneqq \frac{|g|}{\|g\|_{p'}}.$$

By the homogeneity identity (11.3), $\|F\|_p = 1$ and $\|G\|_{p'} = 1$. Applying Young's inequality (Theorem 11.9) at each point $x \in X$ with $a = F(x)$ and $b = G(x)$ yields

$$F(x)G(x) \leq \frac{1}{p}F(x)^p + \frac{1}{p'}G(x)^{p'}.$$

Step 3: Integration and conclusion.
Integrating this pointwise inequality over X with respect to μ gives

$$\int_X FG \, \mathrm{d}\mu \leq \frac{1}{p}\int_X F^p \, \mathrm{d}\mu + \frac{1}{p'}\int_X G^{p'} \, \mathrm{d}\mu = \frac{1}{p}\|F\|_p^p + \frac{1}{p'}\|G\|_{p'}^{p'} = \frac{1}{p} + \frac{1}{p'} = 1.$$

Multiplying both sides of this inequality by the positive product $\|f\|_p \, \|g\|_{p'}$ yields

$$\int_X |fg| \, \mathrm{d}\mu \leq \|f\|_p \, \|g\|_{p'}.$$

Because the right-hand side is finite, $fg \in L^1(X, \mu)$. □

For $p = p' = 2$, Hölder's inequality is the **Cauchy–Schwarz inequality** $\int_X |fg| \, \mathrm{d}\mu \leq \|f\|_2 \|g\|_2$. For the counting measure on $\mathbb{N}$, Hölder's inequality states that for sequences $(a_k) \in \ell^p$ and $(b_k) \in \ell^{p'}$,

$$\sum_{k=1}^{+\infty} |a_k b_k| \leq \left(\sum_{k=1}^{+\infty} |a_k|^p \right)^{1/p} \left(\sum_{k=1}^{+\infty} |b_k|^{p'} \right)^{1/p'}.$$

For the counting measure on $\{1, \dots, n\}$ and $p = 2$, this inequality reduces to the Cauchy–Schwarz inequality for the dot product on $\mathbb{R}^n$.

Remark (The endpoint $p = 1$). The exponent $p = 1$ has no conjugate exponent in $(1, +\infty)$. The role of $L^{p'}$ is then played by bounded functions: if $|g| \leq M$ almost everywhere and $f \in L^1(X, \mu)$, monotonicity of the integral gives $\int_X |fg| \, \mathrm{d}\mu \leq M \|f\|_1$. The corresponding space $L^\infty(X, \mu)$ of essentially bounded functions is treated separately.

11.4 Minkowski's inequality

With Hölder's inequality established, we return to the triangle inequality for $\|\cdot\|_p$ on $L^p(X, \mu)$ formulated in Section 11.1.3.

Theorem 11.11 (Minkowski's inequality). For $p \in [1, +\infty)$ and $f, g \in L^p(X, \mu)$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p. \tag{11.9}$$

The proof decomposes $|f + g|^p$ into the sum $|f| \, |f + g|^{p-1} + |g| \, |f + g|^{p-1}$, applies Hölder's inequality to each product, and divides by $\|f + g\|_p^{p-1}$.

Proof of Theorem 11.11. Step 1: The endpoint $p = 1$ and reduction to positive norm.
For $p = 1$, integrating the pointwise triangle inequality $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ over X yields $\|f + g\|_1 \leq \|f\|_1 + \|g\|_1$. Suppose that $p > 1$. By Section 11.1.1, $f + g \in L^p(X, \mu)$, so $\|f + g\|_p < +\infty$. If $\|f + g\|_p = 0$, the inequality (11.9) holds because the right-hand side is non-negative. We may therefore assume that $\|f + g\|_p > 0$.

Step 2: Pointwise decomposition and integrability of the auxiliary factor.
At every point $x \in X$, the triangle inequality in $\mathbb{R}$ gives

$$|f(x) + g(x)|^p = |f(x) + g(x)| \, |f(x) + g(x)|^{p-1} \leq |f(x)| \, |f(x) + g(x)|^{p-1} + |g(x)| \, |f(x) + g(x)|^{p-1}.$$

We define the auxiliary function $h \colon X \to \mathbb{R}$ by $h(x) = |f(x) + g(x)|^{p-1}$. By the identity $(p - 1)p' = p$ from (11.5), the p'-th power of h satisfies $h^{p'} = |f + g|^{(p-1)p'} = |f + g|^p$. Integrating over X yields

$$\int_X h^{p'} \, \mathrm{d}\mu = \int_X |f + g|^p \, \mathrm{d}\mu = \|f + g\|_p^p < +\infty.$$

Hence $h \in L^{p'}(X,\mu)$, and its $L^{p'}$ norm is

$$\|h\|_{p'} = \left(\int_X h^{p'}\,\mathrm{d}\mu\right)^{1/p'} = \left(\|f+g\|_p^p\right)^{1/p'} = \|f+g\|_p^{p/p'} = \|f+g\|_p^{p-1},$$

where the last equality uses $p/p' = p-1$ from (11.5).

Step 3: Application of Hölder's inequality and conclusion.
Applying Hölder's inequality (Theorem 11.10) to the pairs $(|f|,h)$ and $(|g|,h)$ gives

$$\int_X |f|h\,\mathrm{d}\mu \le \|f\|_p\,\|h\|_{p'}, \qquad \int_X |g|h\,\mathrm{d}\mu \le \|g\|_p\,\|h\|_{p'}.$$

Integrating the pointwise decomposition from Step 2 and combining these bounds yields

$$\|f+g\|_p^p = \int_X |f+g|^p\,\mathrm{d}\mu \le \int_X |f|h\,\mathrm{d}\mu + \int_X |g|h\,\mathrm{d}\mu \le (\|f\|_p + \|g\|_p)\|h\|_{p'} = (\|f\|_p + \|g\|_p)\|f+g\|_p^{p-1}.$$

Because $\|f+g\|_p$ is finite and strictly positive, dividing both sides by $\|f+g\|_p^{p-1}$ yields (11.9). □

Corollary 11.12 (L^p is a normed space). For $p \in [1,+\infty)$, the function $\|\cdot\|_p$ is a norm on the vector space $L^p(X,\mu)$: for all $f,g \in L^p(X,\mu)$ and $\lambda \in \mathbb{R}$,

$$\|f\|_p = 0 \iff f = 0, \qquad \|\lambda f\|_p = |\lambda|\,\|f\|_p, \qquad \|f+g\|_p \le \|f\|_p + \|g\|_p.$$

Proof of Corollary 11.12. We verify the three norm axioms for $\|\cdot\|_p$ on $L^p(X,\mu)$. Point separation $\|f\|_p = 0 \iff f = 0$ in $L^p(X,\mu)$ was established in Definition 11.5; absolute homogeneity $\|\lambda f\|_p = |\lambda|\,\|f\|_p$ holds for all $\lambda \in \mathbb{R}$ by (11.3); and subadditivity is Minkowski's inequality (Theorem 11.11). □

Consequently, the distance $d_p(f,g) \coloneqq \|f-g\|_p$ defines a metric on $L^p(X,\mu)$, and a sequence $(f_n)_{n\in\mathbb{N}}$ converges to f in $L^p(X,\mu)$ if $\|f_n - f\|_p \to 0$ as $n \to \infty$. For $p = 1$, this is the convergence in L^1 that appears in the conclusion of the dominated convergence theorem (Note 22).

11.5 Exponents below one: a quasinorm

The definitions of the space $\mathcal{L}^p(X,\mu)$, the functional $\|\cdot\|_p$, and the quotient space $L^p(X,\mu)$ in Definition 11.2 and Definition 11.5 remain formally unchanged for exponents $p \in (0,1)$. The proof that $\mathcal{L}^p(X,\mu)$ is a vector space relied only on the crude bound (11.4), which holds for every $p > 0$. Similarly, absolute homogeneity (11.3) and the separation property $\|f\|_p = 0 \iff f = 0$ in $L^p(X,\mu)$ hold for all $p \in (0,+\infty)$. However, the proof of Minkowski's inequality required a conjugate exponent $p' > 1$, which exists only when $p > 1$. When $p \in (0,1)$, the triangle inequality fails.

Example 11.13 (Failure of the triangle inequality for $p < 1$). On $[0,1]$ equipped with Lebesgue measure, consider the indicator functions $f \coloneqq \mathbb{1}_{[0,1/2]}$ and $g \coloneqq \mathbb{1}_{(1/2,1]}$, so that $f + g = \mathbb{1}_{[0,1]}$. For an exponent $p \in (0,1)$,

$$\|f\|_p = \|g\|_p = \left(\tfrac{1}{2}\right)^{1/p} = 2^{-1/p}, \qquad \|f+g\|_p = 1.$$

Because $p \in (0,1)$, we have $1 - 1/p < 0$, so the sum of norms evaluates to

$$\|f\|_p + \|g\|_p = 2 \cdot 2^{-1/p} = 2^{1-1/p} < 1 = \|f+g\|_p.$$

Thus the triangle inequality fails for these functions.

In Definition 11.3, we introduced the notion of a quasinorm as a functional satisfying point separation, absolute homogeneity, and the quasi-triangle inequality $\|u+v\| \le C(\|u\|+\|v\|)$ for some constant $C \ge 1$. While the elementary bound established in Section 11.1.1 showed that $\|\cdot\|_p$ is a quasinorm on $L^p(X,\mu)$ with constant $C \le 2^{1+1/p}$, Example 11.13 reveals that the ratio $\|f+g\|_p/(\|f\|_p+\|g\|_p)$ in that instance equals $2^{1/p-1}$. We now show that $C_p = 2^{1/p-1}$ is the exact, optimal quasinorm constant for every exponent $p \in (0,1)$.

Theorem 11.14 ($\|\cdot\|_p$ is a quasinorm). For $p \in (0,+\infty)$, the function $\|\cdot\|_p$ is a quasinorm on $L^p(X,\mu)$. For all $f, g \in L^p(X,\mu)$,

$$\|f+g\|_p \le C_p(\|f\|_p + \|g\|_p), \qquad C_p \coloneqq \begin{cases} 1, & p \ge 1, \\ 2^{1/p-1}, & 0 < p < 1. \end{cases} \tag{11.10}$$

The proof for $p \in (0,1)$ proceeds in two steps. First, because $t \mapsto t^p$ is subadditive on $[0,+\infty)$ when $p < 1$, the power functional $\|\cdot\|_p^p$ satisfies the triangle inequality without a constant. Second, raising both sides to the power $1/p > 1$ introduces the constant factor $2^{1/p-1}$ by the convexity of $u \mapsto u^{1/p}$.

Proof of Theorem 11.14. Point separation $\|f\|_p = 0 \iff f = 0$ in $L^p(X,\mu)$ and absolute homogeneity $\|\lambda f\|_p = |\lambda|\,\|f\|_p$ were established in Section 11.1.2 and (11.3) for all $p > 0$. For $p \ge 1$, the inequality (11.10) with $C_p = 1$ is Minkowski's inequality (Theorem 11.11). Suppose that $p \in (0,1)$.

Step 1: Subadditivity of the p-th power functional.
We claim that $(a+b)^p \le a^p + b^p$ for all $a, b \ge 0$. If $a+b=0$, both sides vanish. Suppose that $a+b>0$. The ratios $u \coloneqq a/(a+b)$ and $v \coloneqq b/(a+b)$ lie in $[0,1]$ and satisfy $u+v=1$. Because $p \in (0,1)$, the function $t \mapsto t^p$ satisfies $t^p \ge t$ for all $t \in [0,1]$. Therefore

$$\left(\frac{a}{a+b}\right)^p + \left(\frac{b}{a+b}\right)^p \ge \frac{a}{a+b} + \frac{b}{a+b} = 1.$$

Multiplying both sides by $(a+b)^p$ proves the claim. Applying this inequality pointwise with $a = |f(x)|$ and $b = |g(x)|$, and using the numerical triangle inequality $|f(x)+g(x)| \le |f(x)| + |g(x)|$, we integrate over X to obtain

$$\|f+g\|_p^p \le \|f\|_p^p + \|g\|_p^p. \tag{11.11}$$

Step 2: Conversion to $\|\cdot\|_p$ via convexity.
We define $r \coloneqq 1/p > 1$. We claim that $(s+t)^r \le 2^{r-1}(s^r+t^r)$ for all $s, t \ge 0$. If $s=0$ or $t=0$, the inequality holds because $2^{r-1} \ge 1$. If $s>0$ and $t>0$, the convexity of $u \mapsto u^r$ on $(0,+\infty)$ with weight $1/2$ gives

$$\left(\frac{s+t}{2}\right)^r \le \frac{1}{2}(s^r+t^r).$$

Multiplying both sides by 2^r yields $(s+t)^r \le 2^{r-1}(s^r+t^r)$, proving the claim. Applying this inequality with $s = \|f\|_p^p$ and $t = \|g\|_p^p$ to the subadditivity bound (11.11) raised to the power $r = 1/p$ yields

$$\|f+g\|_p = (\|f+g\|_p^p)^{1/p} \le (\|f\|_p^p + \|g\|_p^p)^{1/p} \le 2^{1/p-1}(\|f\|_p + \|g\|_p).$$

This completes the proof of (11.10). □

Because the functions $f = \mathbb{1}_{[0,1/2]}$ and $g = \mathbb{1}_{(1/2,1]}$ from Example 11.13 satisfy $\|f+g\|_p = 2^{1/p-1}(\|f\|_p + \|g\|_p)$, the constant $C_p = 2^{1/p-1}$ in (11.10) is optimal. Furthermore, the subadditivity bound (11.11) shows that

for $p \in (0,1)$, the quantity $d_p(f,g) \coloneqq \|f-g\|_p^p$, without the root $1/p$, satisfies the triangle inequality and defines a translation-invariant metric on $L^p(X,\mu)$.

11.6 Comparing L^p spaces of different exponents

The integrability of power functions in Example 11.1 showed that $x \mapsto x^{-a}$ belongs to $L^p(0,1]$ for $p \in (0, 1/a)$ and to $L^p[1,+\infty)$ for $p \in (1/a, +\infty)$. On the finite measure space $(0,1]$, membership in L^q implies membership in L^p whenever $p < q$. Using Hölder's inequality, we prove that this inclusion holds on every finite measure space.

Theorem 11.15 (Inclusion of L^p spaces on finite measure spaces). Suppose that $\mu(X) < +\infty$ and $0 < p < q < +\infty$. Then $L^q(X,\mu) \subseteq L^p(X,\mu)$, and for every $f \in L^q(X,\mu)$,

$$\|f\|_p \le \mu(X)^{\frac{1}{p}-\frac{1}{q}} \, \|f\|_q. \tag{11.12}$$

The proof applies Hölder's inequality to the product of $|f|^p$ and the constant function 1 with the exponent $r \coloneqq q/p > 1$. This exponent is chosen so that the r-th power of $|f|^p$ equals $|f|^q$, whose integral is finite by assumption. The constant function 1 belongs to $L^{r'}$ precisely because $\mu(X) < +\infty$.

Proof of Theorem 11.15. We define $r \coloneqq q/p \in (1,+\infty)$, whose conjugate exponent is $r' = \frac{r}{r-1} = \frac{q}{q-p}$. For $f \in L^q(X,\mu)$, the function $|f|^p$ belongs to $L^r(X,\mu)$ with norm $\||f|^p\|_r = \|f\|_q^p$. Indeed,

$$\int_X \left(|f|^p\right)^r \, \mathrm{d}\mu = \int_X |f|^q \, \mathrm{d}\mu < +\infty.$$

Because $\mu(X) < +\infty$, the constant function 1 belongs to $L^{r'}(X,\mu)$, and its $L^{r'}$ norm is $\|1\|_{r'} = \mu(X)^{1/r'} = \mu(X)^{1-p/q}$. Applying Hölder's inequality (Theorem 11.10) to the product $|f|^p \cdot 1$ with conjugate exponents r and r' gives

$$\int_X |f|^p \, \mathrm{d}\mu = \int_X |f|^p \cdot 1 \, \mathrm{d}\mu \le \||f|^p\|_r \, \|1\|_{r'} = \|f\|_q^p \, \mu(X)^{1-\frac{p}{q}} < +\infty.$$

Thus $f \in L^p(X,\mu)$. Raising both sides of this inequality to the power $1/p$ yields (11.12). □

On a finite measure space, a function can fail to be p-integrable only through the growth of its large values; the q-th power with $q > p$ penalizes large values more strongly than the p-th power. When $\mu(X) = 1$, the inequality (11.12) reads $\|f\|_p \le \|f\|_q$, so the norm $\|f\|_p$ is a non-decreasing function of p. The inclusion $L^q(X,\mu) \subseteq L^p(X,\mu)$ is strict in general: for any $q > 0$, the power function $x \mapsto x^{-1/q}$ on $(0,1]$ belongs to $L^p(0,1]$ for every $p < q$ by Example 11.1, but it does not belong to $L^q(0,1]$.

Warning (Failure of inclusion on spaces of infinite measure). The hypothesis $\mu(X) < +\infty$ cannot be omitted from Theorem 11.15. On $\mathbb{R}$ equipped with Lebesgue measure $\mathcal{L}$, the inclusion $L^q(\mathbb{R}) \subseteq L^p(\mathbb{R})$ fails in general. For instance, setting $a = 1$ in Example 11.1 shows that the function $f(x) \coloneqq x^{-1}\mathbb{1}_{[1,+\infty)}(x)$ belongs to $L^2(\mathbb{R})$ but not to $L^1(\mathbb{R})$. Conversely, setting $a = 1/2$ shows that the function $g(x) \coloneqq x^{-1/2}\mathbb{1}_{(0,1]}(x)$ belongs to $L^1(\mathbb{R})$ but not to $L^2(\mathbb{R})$. Consequently, neither of the spaces $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$ contains the other.

On $[1,+\infty)$, the exponents for which x^{-a} is p-integrable form the interval $(1/a, +\infty)$, so membership in L^p implies membership in L^q for larger exponents q. An inclusion from smaller to larger exponent spaces holds on measure spaces where positive measures cannot be arbitrarily small, such as $\mathbb{N}$ with counting measure, where every non-empty set has measure at least 1.

Problem 11.16 (Inclusion when positive measures are bounded below). Suppose that there exists a constant $m > 0$ such that every measurable set $A \in \mathcal{A}$ with $\mu(A) > 0$ satisfies $\mu(A) \geq m$, and that $0 < p < q < +\infty$. Prove that $L^p(X,\mu) \subseteq L^q(X,\mu)$ and that, for every $f \in L^p(X,\mu)$,

$$\|f\|_q \leq m^{\frac{1}{q}-\frac{1}{p}} \, \|f\|_p.$$

Deduce that $\ell^p \subseteq \ell^q$ with $\|a\|_{\ell^q} \leq \|a\|_{\ell^p}$ for sequence spaces.
Hint. By Markov's inequality, the set $\{x \in X : |f(x)| > t\}$ has measure strictly less than m whenever $t > m^{-1/p}\|f\|_p$. Because every measurable set of positive measure satisfies $\mu(A) \geq m$, this implies that $\mu(\{|f| > t\}) = 0$ for all such t, so $|f| \leq m^{-1/p}\|f\|_p$ almost everywhere. The bound on $\|f\|_q$ then follows by decomposing $|f|^q = |f|^{q-p}|f|^p$ and integrating.

11.7 Hölder's inequality and duality

Beyond estimating product integrals, Hölder's inequality establishes a fundamental connection between $L^p(X,\mu)$ and $L^{p'}(X,\mu)$ through the theory of duality. Specifically, each function $g \in L^{p'}(X,\mu)$ defines a continuous linear functional on $L^p(X,\mu)$ whose operator norm coincides with $\|g\|_{p'}$.

Definition 11.17 (Bounded linear functional and dual norm). For $p \in [1,+\infty)$, a **bounded linear functional** on $L^p(X,\mu)$ is a linear map $\Lambda : L^p(X,\mu) \to \mathbb{R}$ for which there exists a constant $C \geq 0$ such that

$$|\Lambda(f)| \leq C\|f\|_p \qquad \text{for all } f \in L^p(X,\mu).$$

The set of bounded linear functionals on $L^p(X,\mu)$ is denoted $(L^p(X,\mu))^*$ and is called the **dual space** of $L^p(X,\mu)$. The **dual norm** of $\Lambda \in (L^p)^*$ is

$$\|\Lambda\|_{(L^p)^*} := \sup\{|\Lambda(f)| : f \in L^p(X,\mu),\ \|f\|_p \leq 1\},$$

which is the smallest constant C for which the bound holds.

For a linear map Λ, the bound $|\Lambda(f)| \leq C\|f\|_p$ is equivalent to continuity:

$$|\Lambda(f) - \Lambda(h)| = |\Lambda(f-h)| \leq C\|f-h\|_p,$$

so convergence $f_n \to f$ in $L^p(X,\mu)$ implies $\Lambda(f_n) \to \Lambda(f)$ in $\mathbb{R}$.

For $p \in (1,+\infty)$ and $g \in L^{p'}(X,\mu)$, we define the linear functional $\Lambda_g : L^p(X,\mu) \to \mathbb{R}$ by

$$\Lambda_g(f) := \int_X fg \, \mathrm{d}\mu, \qquad f \in L^p(X,\mu). \tag{11.13}$$

By Hölder's inequality (Theorem 11.10), the product fg is integrable, so $\Lambda_g(f)$ is a well-defined real number. The value does not depend on the choice of representatives: if $f \sim \tilde{f}$ and $g \sim \tilde{g}$, then $fg \sim \tilde{f}\tilde{g}$, and their integrals coincide. Because the Lebesgue integral is linear on $L^1(X,\mu)$ (Note 21), the map $f \mapsto \Lambda_g(f)$ is linear. Furthermore, Hölder's inequality gives

$$|\Lambda_g(f)| \leq \int_X |fg| \, \mathrm{d}\mu \leq \|g\|_{p'} \, \|f\|_p \qquad \text{for all } f \in L^p(X,\mu),$$

which proves that $\Lambda_g \in (L^p)^*$ with operator norm $\|\Lambda_g\|_{(L^p)^*} \leq \|g\|_{p'}$. The next proposition proves that this upper bound is an equality.

Proposition 11.18 (Elements of $L^{p'}$ as functionals on L^p). For $p \in (1, +\infty)$ and $g \in L^{p'}(X, \mu)$, the functional Λ_g defined in (11.13) satisfies

$$\|\Lambda_g\|_{(L^p)^*} = \|g\|_{p'}.$$

Consequently, the map $g \mapsto \Lambda_g$ from $L^{p'}(X, \mu)$ to $(L^p(X, \mu))^*$ is an isometric linear embedding; in particular, distinct elements of $L^{p'}$ define distinct functionals.

Proof of Proposition 11.18. We leave this as an exercise for the reader □

Proposition 11.18 shows that the map $g \mapsto \Lambda_g$ is an isometric embedding of $L^{p'}(X, \mu)$ into the dual space $(L^p(X, \mu))^*$. In particular, the size of g as a function, measured by the $L^{p'}$ norm, equals its operator norm as a continuous linear functional on $L^p(X, \mu)$. Because the relation $1/p + 1/p' = 1$ is symmetric with $(p')' = p$, reversing the roles of f and g provides a dual formula for the L^p norm: for every $f \in L^p(X, \mu)$,

$$\|f\|_p = \max\left\{ \int_X fg \, d\mu : g \in L^{p'}(X, \mu),\ \|g\|_{p'} \le 1 \right\},$$

where the maximum is attained by $g_0 \coloneqq \operatorname{sgn}(f)|f|^{p-1}/\|f\|_p^{p-1}$ when $\|f\|_p > 0$. The L^p norm of a function can therefore be evaluated by testing against elements in the unit ball of $L^{p'}(X, \mu)$.

For the counting measure on $\{1, \dots, n\}$, the space L^p is $\mathbb{R}^n$ equipped with the norm $\|a\|_p = \left(\sum_{k=1}^n |a_k|^p\right)^{1/p}$, and $\Lambda_b(a) = \sum_{k=1}^n a_k b_k$ is the standard dot product with a fixed vector b. Every linear functional on $\mathbb{R}^n$ is a dot product with a unique vector b, and Proposition 11.18 identifies its dual norm as $\|b\|_{p'}$. On a general measure space and for $p \in (1, +\infty)$, every bounded linear functional on $L^p(X, \mu)$ is of the form Λ_g for some $g \in L^{p'}(X, \mu)$; this surjectivity theorem is the Riesz representation theorem for L^p, whose proof we will see in later courses.